

Henry D. Smith

Provo, UT • (385) 489-4215 • hsmith40@byu.edu • [linkedin.com/in/henry-smith-816a72335/](https://www.linkedin.com/in/henry-smith-816a72335/)

EDUCATION

Brigham Young University

Fall 2023 – Present
Provo, UT

- Russell M. Nelson Scholarship recipient
- Major: Mechanical Engineering. Minors: Computer Science, Mathematics.
- Current GPA: 4.0
- Expected Graduation Date: April 2028
- Relevant Coursework: System Dynamics, Mechatronics, Data Structures, Numerical Methods, Engineering Design (CAD)

RESEARCH & ENGINEERING EXPERIENCE

Lawrence Livermore National Laboratory

Summer 2026

Robotics and Controls Intern, LIDSS Group

Livermore, CA

- Developed a hardware-in-the-loop interface between an autonomous raft control board and ShipSim3 using Python, UDP, and serial communication, enabling closed-loop controller testing without deploying a physical raft
- Designed a multithreaded communication architecture that achieved UDP forwarding below 1 ms and 95th-percentile serial latency of 21 ms, satisfying a 50 ms real-time requirement
- Identified a three-degree-of-freedom hydrodynamic damping model from wet-test GPS, heading, and motor-command data using Savitzky-Golay filtering and linear least-squares regression
- Validated the simulated raft against wet-test and independent mission data, achieving 15% yaw-rate NRMSE on wet-test data and 22% yaw-rate NRMSE on mission data

BYU MAGICC Lab

Sept 2026 - Present

Research Assistant

Provo, UT

- Developing a VTOL extension to the ROSflight autopilot framework by integrating fixed-wing and multicopter control architectures
- Working with ROS 2, C++, flight-control software, simulation, and autonomous UAV systems

BYU Robotics and Dynamics Lab

Fall 2025

Research Assistant

Provo, UT

- Deployed an existing ROS 1 control stack on a second mobile robot, adapting configuration, calibration, and launch files for multi-robot operation
- Tested closed-loop robot behavior with external position feedback and diagnosed integration issues across sensing, control, and hardware nodes

BYU Compliant Mechanisms and Robotics Lab

Summer 2024

Research Assistant

Provo, UT

- Supported a graduate-led design effort for the NASA BIG Idea Challenge by assisting with fabrication, assembly, and testing of a roller mechanism

PROJECTS

BYU Mars Rover Capstone Team

Winter 2026

Software and Autonomy Volunteer

Provo, UT

- Redesigned the rover control interface's image-retrieval workflow to improve the reliability of camera-image retrieval during the autonomy task

Line-Following Autonomous Robot (ME EN 330)

- Programmed an embedded-C mobile robot to follow a line and avoid obstacles using real-time sensor feedback, motor-control logic, and calibrated reflectance/distance sensors

TECHNICAL SKILLS

Programming Languages: Python, C++, Embedded C

CAD and Design: SolidWorks, 3D printing, mechanical prototyping

Software Tools: Git, GitHub, ROS 1, ROS 2, Linux (Ubuntu), NumPy, SciPy, Matplotlib
